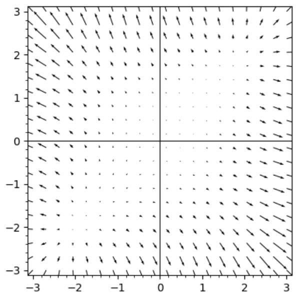
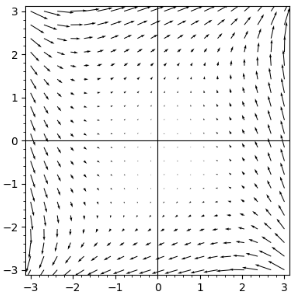
Complex Variable - Midterm Mock Exam

考試時間 Test time 10:30:00 am to 12:00:00 pm; 1.5 hours. 以教室投影機時鐘為準 Using the projected clock. 用空白紙張作答 Answer on blank paper. 每個問號一分 One point per question mark. 全對才給分 No partial credit. 可以帶 991 You can use fx-991 or anything weaker. 舉手問問題 Raise your hand to ask questions.

[1] A vector is ?-free if $P_x + Q_y = 0$. Which of the following picture is this-free?

(A) 

(B) 

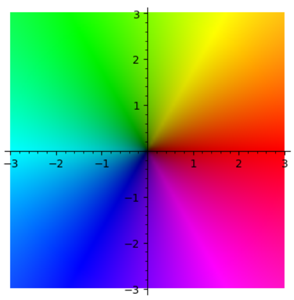
[2] A vector is ?-free if $P_y = Q_x$. Which of the picture above is this-free?

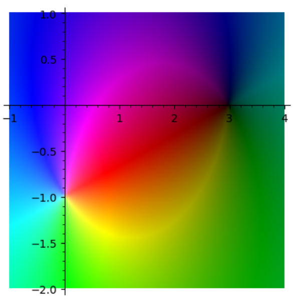
[3] Select all that apply. For $u\colon \mathbf{R}^2 \to \mathbf{R}$ a harmonic function, The vector field (u_x, u_y) is (C) divergence-free (D) curl-free (E) path dependent (F) diffeomorphic?

[4] Consider using the anti-derivative of $P - Qi$ to solve for fluid dynamics problems. Which are correct? (G) fluid flows faster when the Ψ contours are closer. (H) fluid flows faster when the Φ contours are closer. (I) fluid flows faster when the Ψ contours are farther. (J) fluid flows faster when the Φ contours are farther.

[5] What condition is not needed when proving that a subset of $\mathbf{C}$ is $\mathbf{C}$ itself? (K) nonempty (L) open (M) close (N) compact

[6] The lower left picture is the domain coloring plot of $h(z) = z$. The lower picture is $?(? - z)/(z + ?)$. (All question marks are Gaussian integers.)



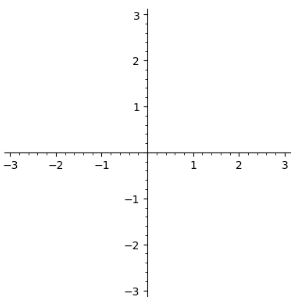


[7] Decompose the Mobius map $d(z-a)/(z-b)$ into to a composition of $+c$, $c\cdot$, and ${}^{-1}$.

$$\mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C} \longrightarrow \mathbf{C}$$

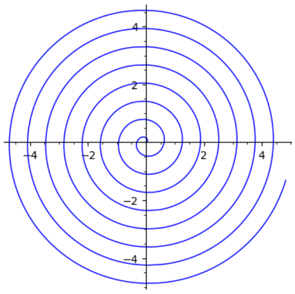
[8] Consider the Mobius map $g(z) = 6/(7 - z)$. Let G be the composition of 10 g's, then $1/G'(1) = 60466???$.

[9] The conformal map $z \mapsto z + 1/z$ maps what to the upper half plane? Draw on the canvas.



[10] $\exp(2000 + i) = ?.??70106084616010^{868} + ?.??59005203013710^{868} i$.

[11] Define a complex $\ln$ function such that $\ln(1) = 0$ and is continuous at points not on the spiral. What is $\ln(3 + 3i)$???



[12] Let f be an arbitrary holomorphic function. The integral $\int_0^\infty \frac{f(x^3)}{1+x^3} dx$ can be expressed using f as ??? (Do not use the integral symbol in your answer.)

[13] Consider an integral of the form $\int_{-1}^{2} f(x)\, dx$. If f is meormorphic and has a pole at $x = 0$, then it probably does not converge. Its (Cauchy) principal value is defined as $\lim_{\delta\to 0} \int_?^? f(x)\, dx + \int_?^? f(x)\, dx$. For $h(x) = \sin(x)/x^4$, the principal value is $0.0???841052008818$.

[14] The Schwarz–Christoffel formula $f(\zeta) = \int_0^\zeta (z+1)^{-?}(z+2)^{-?}\, dz$ maps the upper half plane to a equilateral triangle. The number ? is mapped to the lower left corner. The number ? is mapped to the lower right corner. The number ? is mapped to the top corner. The side length is $?.2999162508563$.

[15] List three (???) bijective conformal maps from the upper half plane to the triangle in the previous problem, and they must be different from the f above. If there are not enough functions, write down NOMORE.

[16] For a harmonic function $u\colon \mathbf{C} \to \mathbf{R}$, there exists another harmonic function $v\colon \mathbf{C} \to \mathbf{R}$ such that $u + iv$ is holomorphic. Let $u(x, y) = -(y^2 - x^2 - 1)/(4y^2x^2 + (y^2 - x^2 - 1)^2)$ and $v(0, 0) = 0$. Then $v(1, 1.5) = -0.???034482758621$.

[17] Consider the Dirichlet problem on a circle. That is, u is harmonic inside the circle, continuously extending to the boundary, and $u(e^{i\theta}) = \beta(\theta)$ for the boundary condition $\beta(\theta) = \cos(\theta)^{72}$. Then $u(0 + 0i) = 0.???770122224943$.

[18] If f and g are holomorphic functions on the entire $\mathbf{C}$ and $f(1/n) = g(1/n)$ for all positive integer n, why is $f = g$ on the entire $\mathbf{C}$? (Tell me the name of the theorem used.) If f and g are holomorphic on the punctured plane $\mathbf{C} \setminus \{0\}$, does $f = g$ still hold or you can find a counterexample???